

RESEARCH

Open Access



Genome-wide analysis of the *BES1* gene family reveals their involvement in grain development of *Triticum aestivum* L

Yang Zhang^{1†}, Yanmin Xu^{1,2†}, Yulu Mao^{1,2}, Min Wang¹, Xiaoli Li¹, Lanfang Jiang¹, Jianyu Hao¹, Dingyi Zhang¹, Hutai Ji^{1*} and Xiaofei Ma^{1*}

Abstract

Background The BRI1-EMS SUPPRESSOR1 (*BES1*) gene family was initially recognized as specifically regulating brassinosteroids to mediate gene expression, which is of vital significance for plant growth and enhancing stress tolerance. Despite extensive studies in multiple plants, there has been a lack of focused and systematic analysis of *BES1s* in wheat grains.

Results In this study, we performed a comprehensive bioinformatics analysis of the *BES1s* in wheat, utilizing the latest genomics data from the Chinese Spring. A total of 19 *TaBES1* were identified. An analysis of conserved domains, phylogenetic relationships, and gene structure revealed a significant level of conservation among *TaBES1s*. A gene collinearity analysis indicated that fragment duplication was the primary mechanism responsible for the amplification of *TaBES1s*. Furthermore, *cis*-acting elements within the promoters of *TaBES1s* were found to be implicated in grain development. Subsequently, SNP analysis revealed the genetic variation of *TaBES1s* across different wheat varieties. Moreover, published RNA-seq data were used, and RNA-seqs of Yaomai36, Pinyu8175, Pinyu8155, and Yaomai30 were performed to identify *TaBES1s* influencing grain development. Finally, the research found that *TaBES1s* had no self-activating activity in wheat. However, the interacting proteins of *TaBES1-1* and *TaBES1-4* are not only involved in starch metabolism but may also be implicated in cell signal transduction.

Conclusions This study further confirmed the potential function of *BES1s* in the grain development of wheat. These findings that *BES1s* play a regulatory role in wheat grain development provide a foundation for further understanding the molecular mechanisms underlying crop grain development.

Keywords Wheat, *BES1* gene family, Grain development, Expression analysis, Bioinformatics analysis

[†]Yang Zhang and Yanmin Xu contributed equally to this work.

*Correspondence:

Hutai Ji

sjjihut@163.com

Xiaofei Ma

nongxue06123@163.com

¹Wheat Research Institute, Shanxi Agricultural University, Linfen 041000, China

²Agricultural College, Shanxi Agricultural University, Jinzhong 030810, China



Introduction

Brassinosteroids [1] represent a class of steroid hormones in plants that are crucial for nearly all physiological processes throughout the plant life cycle [2]. BRI1-EMS SUPPRESSOR1 (*BES1*) is a plant-specific transcription factor and a founding member of the basic helix-loop-helix (bHLH) transcription factor family, which is vital for regulating plant growth and improving plant stress resistance [3]. Research has demonstrated that *BES1* significantly influences BR signal transduction and plays an important role in the signal transduction of abscisic acid (ABA) and strigolactone (SL) [4]. The *BES1* protein contains a putative nuclear localization signal (NLS), a highly conserved N-terminal domain (N), a BIN2 phosphorylation domain (P), a PEST motif, and a carboxy-terminal domain (C) [5]. The *BES1*-NLS is responsible for contact with DNA, and the N-terminal moiety is responsible for binding to DNA, thus forming a DNA binding domain with an atypical helix-loop-helix structure (HLH) [6]. This domain interacts with E-box (CANNTG) or BR response elements (BRRE, CGT/CG) to modulate the expression of target genes [1].

BES1 was initially identified as a crucial transcription factor that specifically modulates BR response and was part of a six-member family that includes *BES1*, *BZR1*, *BEH1*, *BEH2*, *BEH3*, and *BEH4* [7]. To date, researchers have reported the *BES1* family in several plants, including *Arabidopsis thaliana* [8], *Oryza sativa* [9], *Triticum aestivum* L [10], *Zea mays* [11], *Cucumis sativus* [12], and *Lycopersicon esculentum* [13], in which they identified 8, 6, 23, 10, 6, and 9 *BES1*s, respectively. Previous studies have shown that *BES1*s were implicated in plant growth, development, and stress response, such as cell growth [14], anther development [10], flowering regulation [15], grain development [9], and pathogen defense [16]. Additionally, *BES1*s not only act as a directly regulated transcription factor but also interact with other proteins to regulate target genes. The *BES1*-*TPL*-*HDA19* repressor protein complex governs the epigenetic silencing of *ABI3*, thereby inhibiting ABA signaling output during early seedling development [17]. *BES1* directly interacts with the G-protein β subunit *AGB1* to positively regulate cell elongation [14]. The interaction between *BES1* and *ABI5* significantly inhibited the binding of *ABI5* to the downstream gene promoter region, thus promoting the germination of *Arabidopsis* seeds [18].

Besides *Arabidopsis*, numerous studies have been carried out on the *BES1*s in other species. Studies in rice revealed that *BES1* positively regulated rice grain size, and overexpression of *ZmBES1/BZR1-5* can significantly increase grain size and weight [9]. In wheat, *TaBES1-3B2* and *TaBES1-3D2* could enhance drought tolerance and regulate other development [10]. In maize, the *ZmBES1/BZR1-1* could negatively regulate drought stress [19].

In soybeans, *GmBEHL1* (*AtBES1/BZR1* homolog 1) could regulate the number of soybean root nodules [20]. *MdBES1* positively regulates the expression of *MdMYB88* under cold stress and pathogen attack [21]. In tomatoes, *SIBES1.2*, *SIBES1.5*, *SIBES1.6*, and *SIBES1.9* have synergistic effects in regulating tomato fruit development [13].

Wheat ($2n=6x=42$, AABBDD) is the crop with the largest global cultivation area [22]. Enhancing wheat production and quality is essential for maintaining global food security [23]. The grain-filling stage is a pivotal phase in grain development, playing a crucial role in both yield and quality formation [24]. The filling stage is a biological process characterized by the gradual increase in the seed length, width, height, and volume, driven by the continuous formation of grains and the accumulation of dry matter and water [25]. Therefore, analyzing the progression of wheat grains during this filling stage is vital for optimizing both the yield and quality. The *BES1* gene family has been functionally analyzed and characterized across various plant species. Dezhou W et al. also elucidated the roles of *TaBES1*s in stress resistance and anther development in wheat [2]. Building on this foundation, we extended our research focus to the function of *BES1*s in wheat grain development. Through this investigation, we have established a foundation for a more comprehensive understanding of the multifaceted roles that *BES1*s play in the growth and development of wheat.

In this study, we identified 19 *TaBES1*s in wheat through a genome search method utilizing the latest Chinese Spring genome information. Through bioinformatics analysis, we studied conserved domains, phylogenetic relationships, chromosome localization, physicochemical properties, gene structure, collinearity, and *cis*-promoter elements. Additionally, we investigated the expression profile of *TaBES1*s across different tissues and under various abiotic stress conditions in wheat. Subsequently, we analyzed haplotypes of *TaBES1*s to understand their genetic variation in wheat. RNA-seq and qRT-PCR analyses were performed to further identify the *TaBES1*s affecting the grain development in wheat. This study provides essential candidate genes for understanding wheat grain development.

Results

Conserved domains and basic information of *TaBES1*s

In the present investigation, 19 *TaBES1*s were identified and subsequently renamed *TaBES1-1 A* to *TaBES1-7D*. The conserved domain of *TaBES1*s features a basic helix-loop-helix structure (Fig. 1). The basic region comprises 18 to 20 amino acids, situated at the N-terminal end of the bHLH domains, serving as a DNA binding site, while the HLH region is located at the C-terminal end, relying on the interaction of hydrophobic amino acids to form either homologous or heterodimers of two HLH

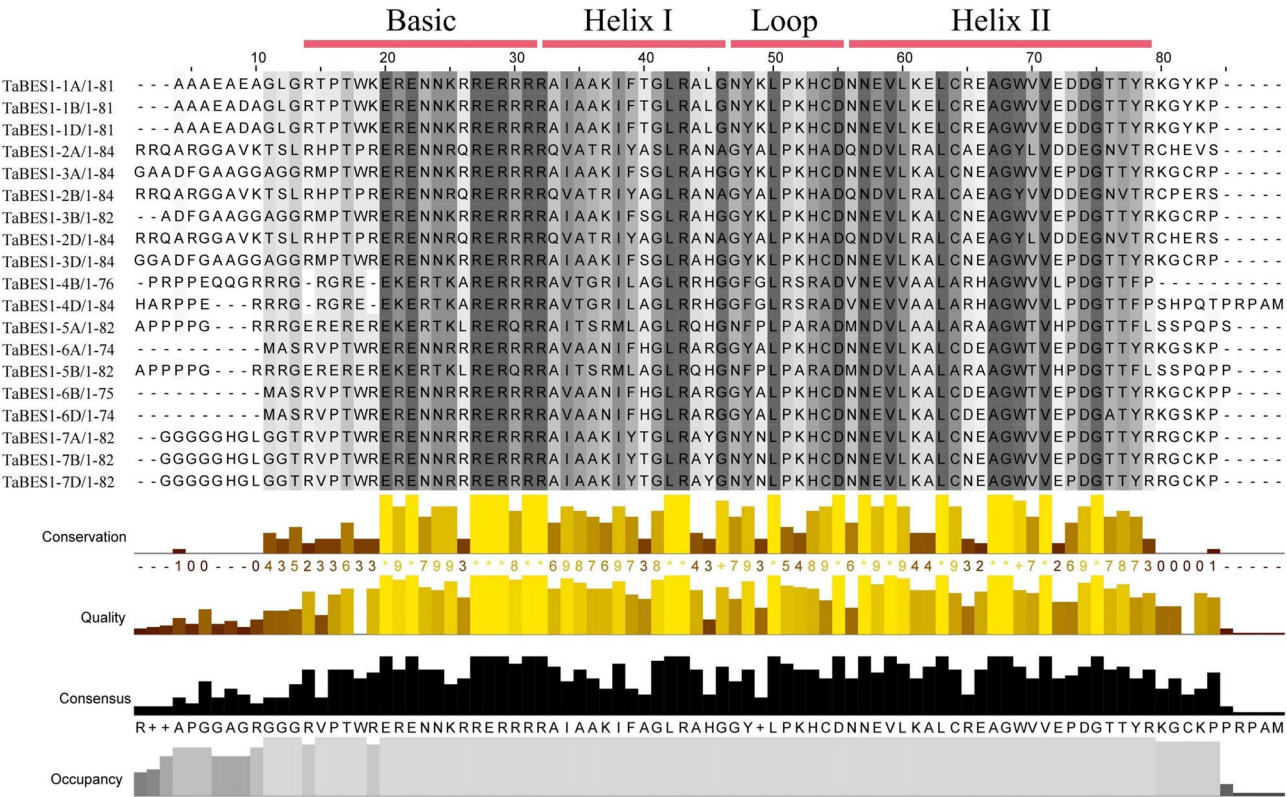


Fig. 1 Conserved-domain of TaBES1 in *Triticum aestivum* L. The red lines represent basic helix-loop-helix conserved domains of TaBES1

proteins [6]. As detailed in Table 1, the length of TaBES1s varied from 178 to 686 amino acids. The CDS sequence length is between 537 and 2061 bp. The molecular weight ranges from 19.27 to 75.48 kDa. The isoelectric point ranges from 5.4 to 9.4. Prediction of subcellular localization revealed that TaBES1s were entirely localized in the nucleus.

Phylogenetic relationships, chromosome distribution, and collinearity analysis of TaBES1s

To investigate the phylogenetic relationship of BES1 in wheat, rice, and Arabidopsis, a phylogenetic evolutionary tree was constructed (Fig. 2A, Table S1). The results indicated that the TaBES1 gene family could be classified into four distinct subfamilies, namely I, II, III, and IV, consistent with classification from previous studies [26]. TaBES1s were distributed across all subfamilies, with subfamily IV containing the highest number of TaBES1s, which contains nine TaBES1s. Subfamily II follows with four TaBES1s, while subfamily I and III each contain three TaBES1s. The BES1s of *A. thaliana* were mainly distributed in subfamily III, while the BES1s of *Oryza sativa* were mainly distributed in subfamily IV. A chromosome localization analysis showed that 19 TaBES1s were unevenly distributed on 14 chromosomes (Fig. 2B). Among them, Chr3A, Chr3B, Chr3D, Chr6A, and Chr6B each contained two TaBES1s, while Chr2A,

Chr2B, Chr2D, Chr4B, Chr4D, Chr6D, Chr7A, Chr7B, and Chr7D each contained one TaBES1.

To further investigate the col-linearity of TaBES1s, the advanced circos plug-in of TBtools was utilized to analyze the col-linearity relationships among TaBES1s. The results indicated that 19 TaBES1s formed 17 pairs of homologous gene pairs, all of which were fragment duplications (Figure S1A), suggesting that fragment duplication was the primary source of TaBES1s amplification. The Ka/Ks ratio, which is the ratio of the non-synonymous mutation rate (Ka) to the synonymous mutation rate (Ks), serves as a valuable metric to reflect the evolutionary patterns of genes. It also allows for the measurement of the type of selective pressures acting on genes during the evolutionary process. In this study, the Ka/Ks values of all homologous gene pairs were Significantly less than 1, indicating that the BES1 gene family has experienced robust purifying selection in wheat throughout its evolutionary history (Table 2).

To further explore the interspecific evolutionary relationships of BES1s, we conducted an analysis of the col-linearity among Arabidopsis, rice, and wheat (Fig. 2C). The results indicated that there were three pairs of col-linear gene pairs between Arabidopsis and wheat, while there were 28 pairs of collinear gene pairs between rice and wheat. This suggests a closer phylogenetic relationship between wheat and rice and also indicates that these

Table 1 Basic information of *BES1s* identified in wheat

Gene name	Gene ID	Chr	Start	End	CDS Length (bp)	Amino acid length(aa)	MW (kDa)	pI	Subcellular localization
TaBES1-1A	TraesCS2A03G0389800	2A	154,748,196	154,750,021	942	313	33.68	8.26	Nucleus
TaBES1-1B	TraesCS2B03G0529800	2B	217,494,348	217,496,101	942	313	33.61	8.26	Nucleus
TaBES1-1D	TraesCS2D03G0418200	2D	153,727,252	153,728,856	942	313	33.63	8.26	Nucleus
TaBES1-2A	TraesCS3A03G0271300	3A	999,892,82	999,899,37	558	185	19.97	9.26	Nucleus
TaBES1-3A	TraesCS3A03G0309300	3A	120,646,485	120,649,266	1071	356	37.41	8.82	Nucleus
TaBES1-2B	TraesCS3B03G0338300	3B	142,067,659	142,068,762	549	182	19.76	9.18	Nucleus
TaBES1-3B	TraesCS3B03G0374100	3B	161,191,528	161,194,198	1065	354	37.28	8.82	Nucleus
TaBES1-2D	TraesCS3D03G0262800	3D	837,533,02	837,561,98	537	178	19.27	9.4	Nucleus
TaBES1-3D	TraesCS3D03G0292400	3D	994,179,24	994,211,22	1077	358	37.51	8.82	Nucleus
TaBES1-4B	TraesCS4B03G0013100	4B	566,577,8	567,002,0	1656	551	62.34	5.4	Nucleus
TaBES1-4D	TraesCS4D03G0010800	4D	338,290,7	338,695,7	2061	686	75.47	5.42	Nucleus
TaBES1-5A	TraesCS6A03G0196400	6A	569,993,28	570,051,26	1962	653	73.23	6.01	Nucleus
TaBES1-6A	TraesCS6A03G0871600	6A	574,557,953	574,561,002	1044	347	36.42	8.79	Nucleus
TaBES1-5B	TraesCS6B03G0285500	6B	107,668,638	107,674,470	2019	672	75.48	6.61	Nucleus
TaBES1-6B	TraesCS6B03G1042400	6B	651,115,225	651,118,180	1071	356	37.72	8.97	Nucleus
TaBES1-6D	TraesCS6D03G0742700	6D	448,521,557	448,524,986	1047	348	36.45	8.62	Nucleus
TaBES1-7A	TraesCS7A03G0867300	7A	523,208,362	523,211,786	1080	359	37.85	8.13	Nucleus
TaBES1-7B	TraesCS7B03G0746500	7B	505,416,429	505,419,594	1080	359	37.86	8.13	Nucleus
TaBES1-7D	TraesCS7D03G0867600	7D	478,688,396	478,692,498	1080	359	37.81	8.13	Nucleus

collinear gene pairs might possess similar biological functions.

Analysis of gene structure and conserved motifs of *TaBES1s*

A gene structure analysis showed that the *TaBES1s* consisted of 2–10 exons and 1–9 introns (Fig. 3A). This finding was consistent with the results for *Lycopersicon esculentum* and *Glycine max Merr* [27]. Among them, subfamilies I, III, and IV each contained two exons, while subfamily II had between 8 and 10 exons, significantly more than the other three subfamilies. This suggests that there might be functional differences among the *TaBES1s*. Except for *TaBES1-2 A*, *TaBES1-4B*, and *TaBES1-4D*, which lack UTR structures, the remaining *TaBES1s* possess a complete gene structure.

The conserved motifs of *TaBES1s* were analyzed using the MEME online tool. The results showed that motif1 and motif2 were more conserved during evolution. They were distributed in all *TaBES1* (Fig. 3B). In addition, the conserved motifs of each subfamily were different. For example, motif8, motif9, and motif10 existed only in subfamily II. Subfamily I contained only motif 1 and motif 2.

Analysis of *cis*-acting elements in the promoter of *TaBES1s*

To gain a more comprehensive understanding of the function of *TaBES1s* in wheat growth and development, we analyzed the *cis*-acting elements of the *TaBES1s* promoters. In total, 18 functional *cis*-acting elements were identified (Fig. 4). 6 *TaBES1s* have GCN4_motif (endosperm expression regulatory element) or RY-element (seed-specific element), indicating that these genes may

be related to grain development (Table S2). Furthermore, MeJA-responsive elements and light-responsive elements were found in all *TaBES1s*.

Haplotype analysis of *TaBES1s*

To further explore the DNA sequence polymorphism of *TaBES1s*, the SNP analysis of 19 *TaBES1s* was performed using Lufei resequencing data. As shown in Table 3, there was no genetic variation in *TaBES1-1D/2A/3A/2B/4B/6D/7A*, suggesting that these genes exhibit relative stability and possess enhanced adaptability across diverse environments. Of the 12 *TaBES1s*, which have genetic variation information, only 3 genes were found to have major haplotypes (*TaBES1-2D/3D/5A*) (Table S3–S5). Among them, *TaBES1-2D/3D* not only showed the same primary haplotype in cultivated varieties, local varieties, and spelt, but also showed the same primary haplotype in different continents as well as in spring and winter wheat. The main haplotype of *TaBES1-5 A* was the same, except in Spelt wheat. Other genes showed different haplotypes in different classifications.

Expression pattern analysis of *TaBES1s* based on RNA-Seq

The tissue-specific expression patterns during grain development of *TaBES1s* were investigated using available RNA-seq data (Table S6). The results indicated that the expression of *TaBES1s* in various tissues of wheat exhibited pronounced tissue-specificity. (Fig. 5A). Most *TaBES1s* were expressed in stamens and anthers, and a few genes were expressed in roots and stems. None of

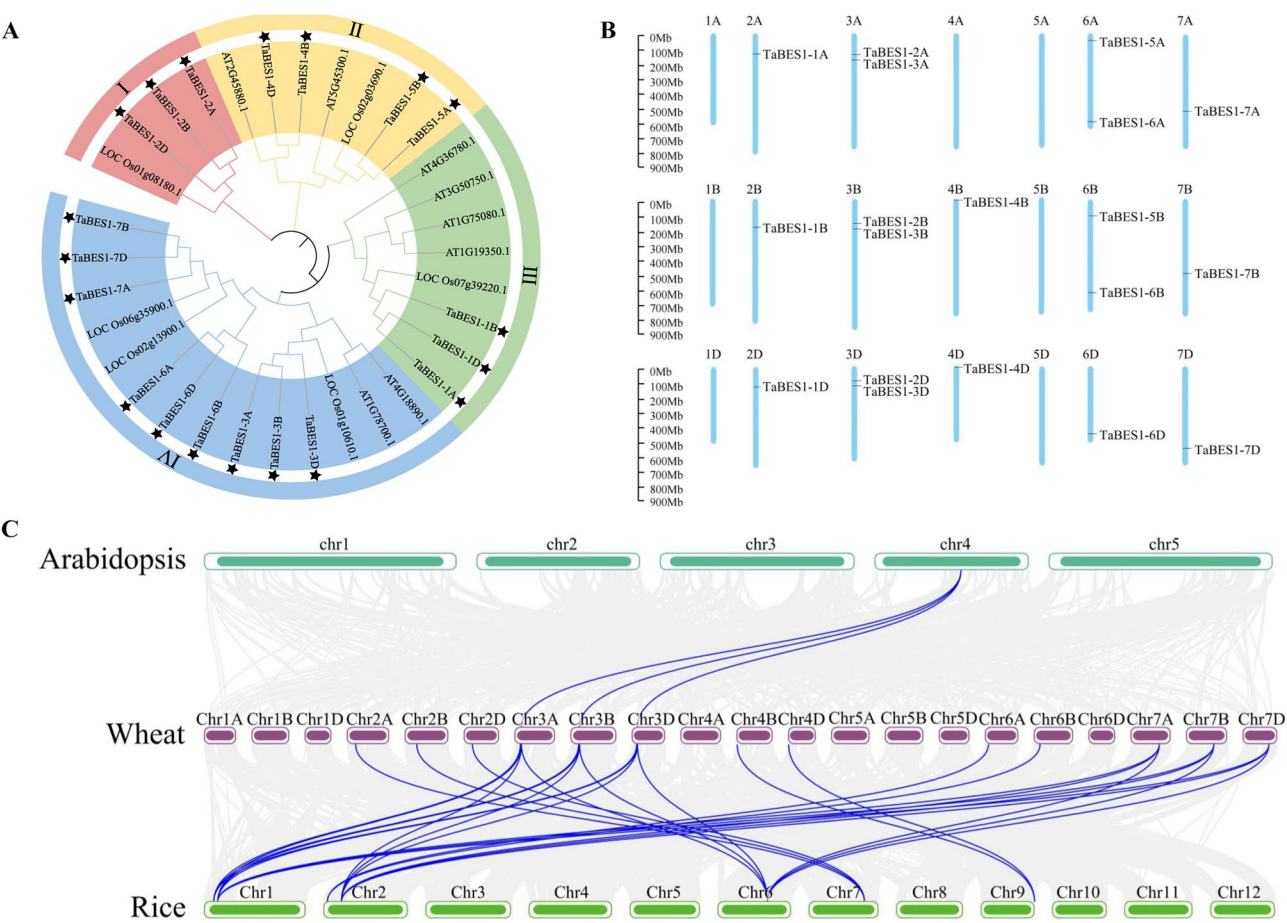


Fig. 2 Analysis of phylogenetic tree, chromosome localization, and collinearity of BES1 gene family in *Triticum aestivum* L. **A** Phylogenetic analysis of BES1 in rice, Arabidopsis, and wheat. Clades I-IV are highlighted in red, yellow, green, and blue, respectively. TaBES1 is marked by a black five-pointed star. **B** Chromosomal distribution of TaBES1s in wheat. The seven chromosome groups with A, B, and D homoeologous chromosomes are indicated. **C** Synteny analysis related to BES1s among the genomes of wheat, Arabidopsis, and rice. The light gray lines represent the collinear blocks between species. The syntenic TaBES1 gene pairs are shown connected by the blue lines

Table 2 Homology analysis of homologous genes of TaBES1s

Homologous genes		Ka	Ks	Ka/Ks	Homologous fragment length	Identities (%)
Gene1	Gene2					
TaBES1-1A	TaBES1-1D	0.005772	0.068816	0.083876	939	94.37
TaBES1-2A	TaBES1-2B	0.065216	0.118693	3	546	91.95
TaBES1-2A	TaBES1-2D	0.058424	0.0972114	0.600998	534	90.88
TaBES1-2B	TaBES1-2D	0	0.12927	0	534	92.28
TaBES1-3A	TaBES1-3B	0	0.195604	0	1062	91.17
TaBES1-3A	TaBES1-3D	0.001252	0.218712	0.005726	1068	92.11
TaBES1-3B	TaBES1-3D	0.039396	0.104517	0.304761	1062	95.32
TaBES1-4B	TaBES1-4D	0.001632	0.154426	0.010566	1614	96.61
TaBES1-5A	TaBES1-5B	0.015872	0.058468	0.271463	1956	98.16
TaBES1-6A	TaBES1-6B	0.043862	0.063215	0.693854	1035	92.2
TaBES1-6A	TaBES1-6D	0.033469	0.077614	0.431226	1035	93.98
TaBES1-6B	TaBES1-6D	0.049481	0.088424	0.559592	1032	90.84
TaBES1-7A	TaBES1-7B	0.006238	0.320583	0.019459	1077	88.63
TaBES1-7A	TaBES1-7D	0.010008	0.255666	0.039146	1077	87.91
TaBES1-7B	TaBES1-7D	0.006238	0.136649	0.045652	1077	93.12

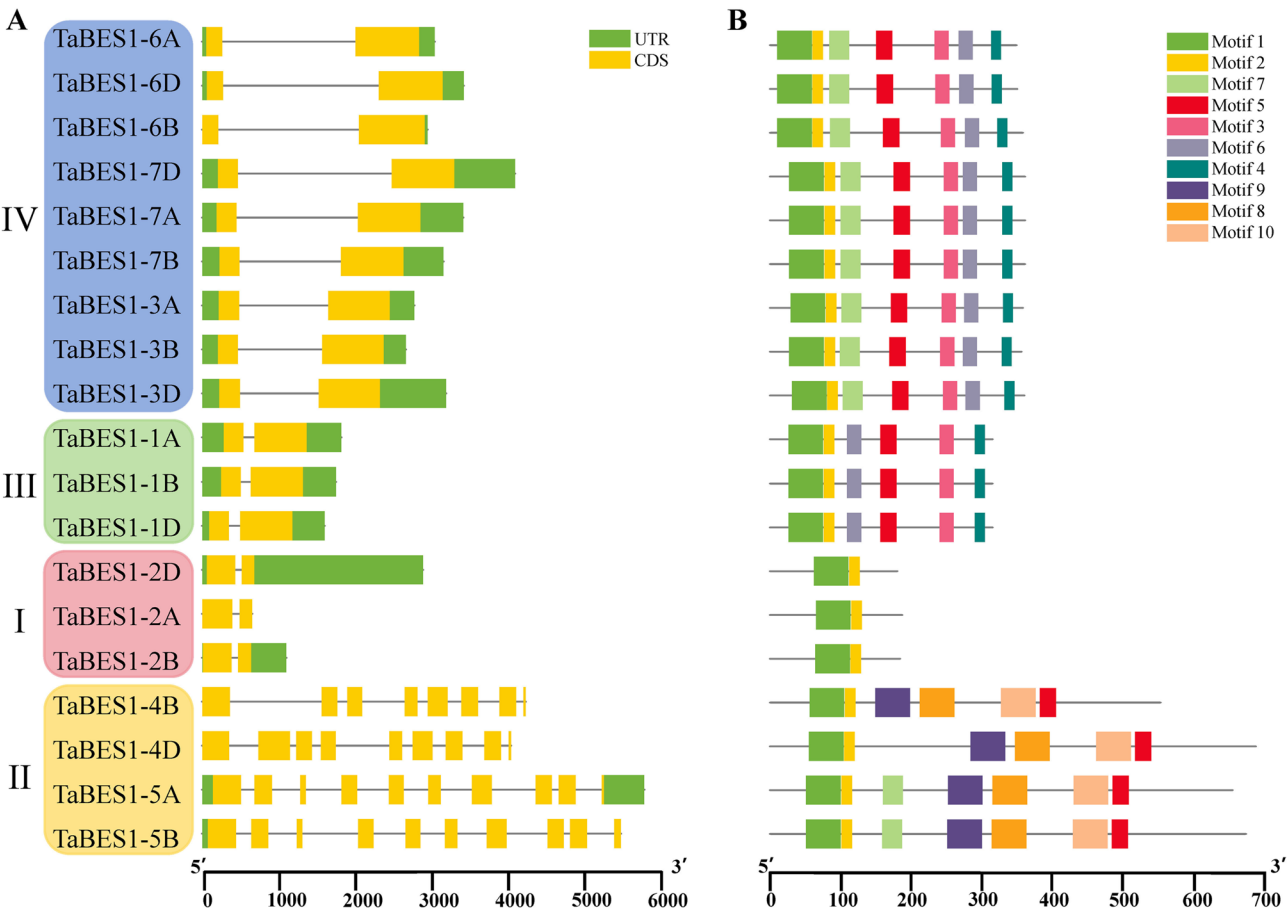


Fig. 3 Analysis of gene structure and motif of TaBES1s. **A** The gene structure of TaBES1s. Green denotes UTRs, whereas yellow suggests CDS. **B** Conserved motif analysis of TaBES1s. 10 identified motifs were represented by different colors. The subfamilies I, II, III, and IV were represented by red, yellow, green, and blue, respectively

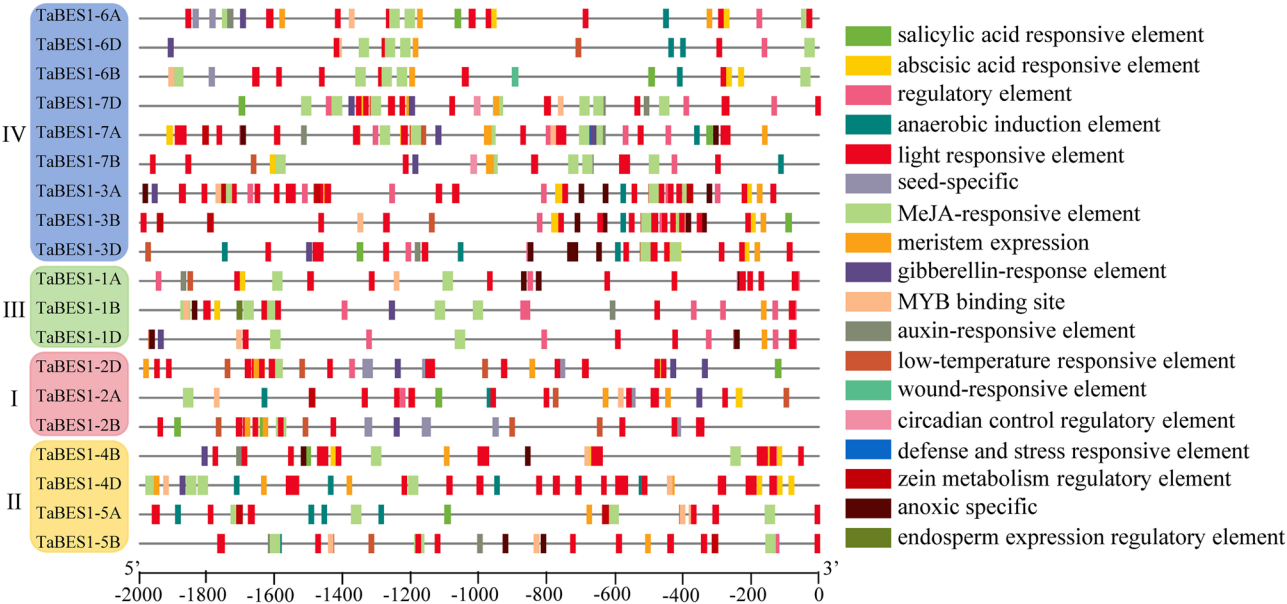


Fig. 4 Cis-acting element analysis of TaBES1s. All cis-elements in promoter regions are listed on the X-axis. The figure indicates the number of cis-elements. The subfamilies I, II, III, and IV were represented by red, yellow, green, and blue, respectively

Table 3 Haplotype number of *TaBES1s*

Gene ID	variety			Geographical location distribution					variety type		
	spelt	cultivate	landrace	Asia	Africa	Europe	N_	S_	Facultative	Spring	Winter
							America	America			
TaBES1-1A	9	9	23	37	6	15	4	3	5	27	23
TaBES1-1B	13	6	15	25	4	18	3	2	4	19	21
TaBES1-1D	0	0	0	0	0	0	0	0	0	0	0
TaBES1-2A	0	0	0	0	0	0	0	0	0	0	0
TaBES1-3A	0	0	0	0	0	0	0	0	0	0	0
TaBES1-2B	0	0	0	0	0	0	0	0	0	0	0
TaBES1-3B	8	5	19	29	4	15	3	3	5	19	19
TaBES1-2D	7	3	8	17	3	9	2	2	5	10	10
TaBES1-3D	3	1	4	6	2	4	1	1	1	6	4
TaBES1-4B	0	0	0	0	0	0	0	0	0	0	0
TaBES1-4D	13	23	83	87	6	37	5	3	10	63	56
TaBES1-5A	5	2	4	5	2	6	1	1	2	4	7
TaBES1-6A	13	6	11	21	4	18	3	3	5	18	18
TaBES1-5B	14	11	26	43	6	24	3	3	8	33	28
TaBES1-6B	14	5	23	35	6	23	4	2	7	28	25
TaBES1-6D	0	0	0	0	0	0	0	0	0	0	0
TaBES1-7A	0	0	0	0	0	0	0	0	0	0	0
TaBES1-7B	14	15	34	47	6	26	4	3	9	34	34
TaBES1-7D	11	2	11	22	5	14	2	1	3	16	16

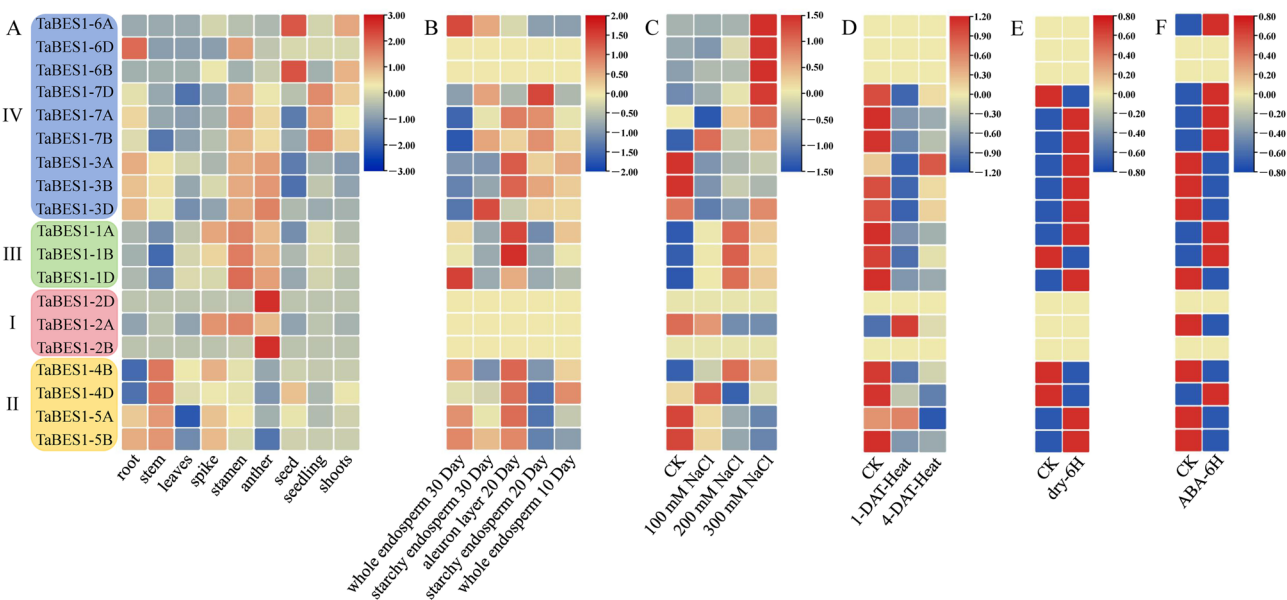


Fig. 5 Expression patterns of *TaBES1s*. **A** A heatmap of *TaBES1s* in root, stem, leaves, spike, stamen, anther, seed, seedling, and shoot. **B** Analysis of expression patterns of *TaBES1s* in endosperm on different days after pollination. The subfamilies I, II, III, and IV were represented by red, yellow, green, and blue, respectively

the *TaBES1s* were expressed in the leaves. To gain deeper insights into the role of *TaBES1s* in grain development, the endosperm from grains at different days post-pollination was analyzed. The results showed except for the *TaBES1-2 A/2B/2D/6B/6D*, others were expressed during grain development (Fig. 5B). Finally, the expression patterns of *TaBES1s* under various abiotic stresses were investigated. The results revealed that *TaBES1s* were

expressed under salt stress, heat stress, drought stress, and ABA stress. For instance, *TaBES16A/6B/6D/7D* exhibited high expression levels under 300 mM salt stress (Fig. 5C). *TaBES12D* was highly expressed after 1 day of heat stress (Fig. 5D). Most *TaBES1s* showed high expression under drought stress (Fig. 5E). The expression of *TaBES16A/7A/7B/7D/1A/1B/4D* was upregulated at 6 h under ABA stress (Fig. 5F). These findings also

corroborate the accuracy of the promoter cis-acting element predictions.

RNA-seq of *TaBES1s* was performed at the grain-filling stage of four winter wheat varieties (Table 4). In Yaomai36, 3 *TaBES1s* were found to be differentially expressed. *TaBES1-1B/4B/6A* had higher expression levels at the milk-ripe stage (MRS). In Pinyu8175, 7 significantly differentially expressed *TaBES1s* were found. *TaBES1-1D/5A/5B/6A* had higher expression levels at MRS, *TaBES1-7D* had higher expression levels at dough period (DP), and *TaBES1-3B/7A* had higher expression levels at wax ripe stage (WRS). In Pinyu8155, 9 significantly differentially expressed *TaBES1s* were found to have higher expression levels at MRS. In Yaomai30, 14 significantly differentially expressed *TaBES1s* were found. *TaBES1-1 A/1B/1D/3B/3D/4B/4D/5B/6A/6B/6D/7B* had higher expression levels at MRS, *TaBES1-7D* had higher

expression levels at DP, and *TaBES1-7 A* had higher expression levels at WRS.

To preliminarily predict the biological functions of the *TaBES1s*, Gene Ontology (GO) analysis was conducted, resulting in the annotation of 89 GO terms, including 79 biological process terms accounting for 89.0%, 5 cell component terms accounted for 5.5%, and 5 molecular function terms accounting for 5.5%, respectively (Figure S1B). In the biological process category, the primary annotation terms were hormone-mediated signaling and responses, gene expression and transcriptional regulation, and cellular responses and metabolism. In the cellular component category, the organelle was a main annotation term. In the molecular function category, transcriptional regulation and signal transduction were the main annotation terms (Table S8).

Table 4 Differently expressed genes associated with grain development in the filling stage

Variety	Gene name	FPKM Mean			P-value	padj	Level
		MRS	DP	WRS			
Yaomai36	TaBES1-1B	5.4051	4.9733	4.024	0.005404657	0.02107149	significant
	TaBES1-4B	7.3702	6.8029	6.5886	0.013570132	0.04376978	significant
	TaBES1-6A	3.1632	1.4913	0.9536	0.000840779	0.00457585	significant
Pinyu8175	TaBES1-1D	4.6364	3.948	3.2956	0.013371519	0.02884624	significant
	TaBES1-3B	6.2493	6.9078	7.1492	0.020881429	0.04266117	significant
	TaBES1-5A	7.4888	6.7591	7.1317	0.008402278	0.01914565	significant
	TaBES1-5B	7.7589	6.8504	7.1497	3.34196×10 ⁻⁵	0.00012703	significant
	TaBES1-6A	3.0604	2.3888	1.6749	0.009388609	0.02112364	significant
	TaBES1-7A	5.259	6.8816	6.973	4.49354×10 ⁻⁹	2.97670×10 ⁻⁸	significant
	TaBES1-7D	6.4604	7.682	7.3566	1.27635×10 ⁻⁷	7.0041378×10 ⁻⁷	significant
Pinyu8155	TaBES1-1A	5.2338	4.3963	3.4308	4.32571×10 ⁻⁵	0.00015333	significant
	TaBES1-1B	4.8893	4.4486	3.6658	0.004829087	0.01113276	significant
	TaBES1-1D	4.9237	3.9552	2.8036	5.87344×10 ⁻⁶	0.00002422	significant
	TaBES1-3A	7.4929	6.754	6.8402	0.022006011	0.04233802	significant
	TaBES1-3B	7.1833	6.7475	6.3991	0.017207485	0.0341517	significant
	TaBES1-3D	7.914	7.2786	7.0936	0.000483798	0.00139865	significant
	TaBES1-4D	8.6491	7.946	8.5448	0.000409169	0.00120139	significant
	TaBES1-5B	7.3965	6.7372	6.767	0.006046495	0.01357849	significant
	TaBES1-6A	2.7867	2.0818	1.2593	0.000842169	0.00231376	significant
Yaomai30	TaBES1-1A	5.9711	5.3473	2.7201	6.96152×10 ⁻¹⁴	4.09925×10 ⁻¹³	significant
	TaBES1-1B	6.0401	5.3714	3.7498	5.68841×10 ⁻¹³	3.13379×10 ⁻¹²	significant
	TaBES1-1D	6.1241	5.0322	3.0933	1.70972×10 ⁻¹⁵	1.12128×10 ⁻¹⁴	significant
	TaBES1-3B	7.3892	7.362	6.9187	0.008926098	0.01621948	significant
	TaBES1-3D	8.0373	7.7436	7.5372	0.002046863	0.0041824	significant
	TaBES1-4B	7.3493	6.6712	6.7377	0.00030284	0.00070403	significant
	TaBES1-4D	8.8213	8.3249	8.5548	0.00674226	0.01254749	significant
	TaBES1-5B	7.1406	6.3297	6.9027	0.014091447	0.02458303	significant
	TaBES1-6A	4.6895	2.9698	1.934	1.85492×10 ⁻¹³	1.057914×10 ⁻¹²	significant
	TaBES1-6B	3.107	1.0852	0.6639	1.28994×10 ⁻⁵	3.58982×10 ⁻⁵	significant
	TaBES1-6D	3.5584	2.7742	2.4175	0.005995873	0.01127078	significant
	TaBES1-7A	6.5554	7.1077	7.1897	0.000144732	0.00035164	significant
	TaBES1-7B	6.7389	6.4616	6.058	0.008617222	0.01570311	significant
	TaBES1-7D	7.5305	7.7134	7.2775	0.024959062	0.04126506	significant

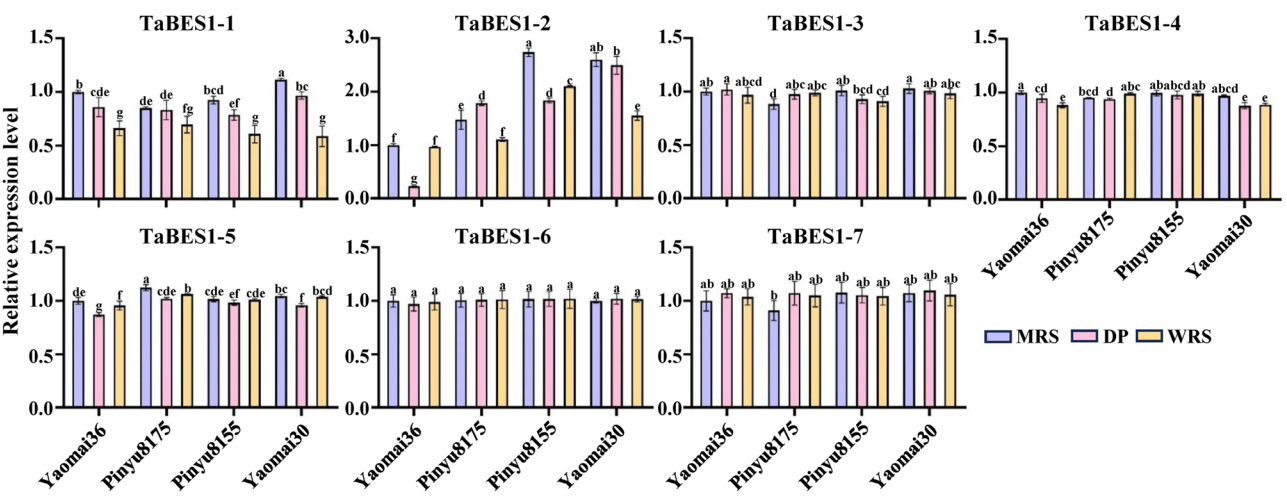


Fig. 6 qRT-PCR verification data analysis of *TaBES1s* in grain development. Each error bar indicates the standard deviation derived from three biological replicates. MRS, DP, and WRS represent the milk ripe stage, dough period, and wax ripe stage, respectively. Different lowercase letters denote statistical significance at the $p < 0.05$ level. *TaActin* was used as a reference gene. The error line represents the standard of three biological replicates

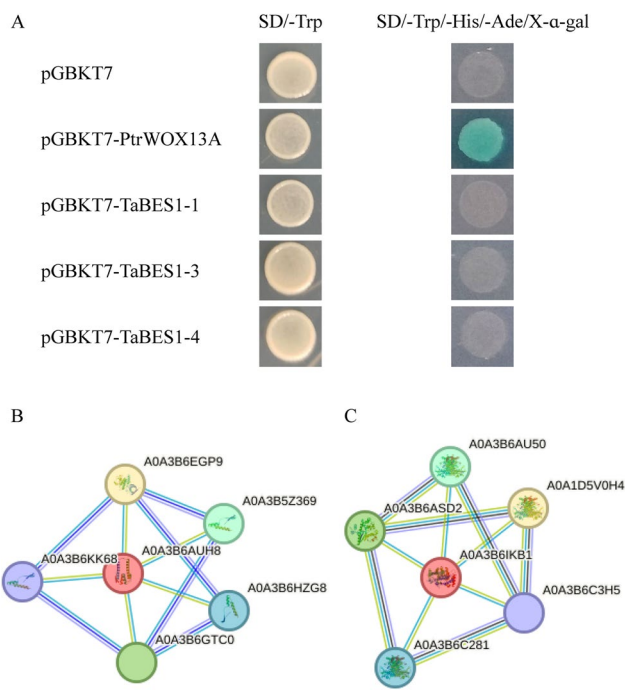


Fig. 7 Auto-activation detection and protein interaction analysis of *TaBES1s*. **A** Transcriptional activation assay of *TaBES1s*. pGBKT7 blank vector and pGBKT7-PtrWOX13A as positive control and negative control, respectively. **B** Interaction network of *TaBES1-1* protein. **C** Interaction network of *TaBES1-4* protein. The nodes in the network represent different proteins, and the lines represent the predicted protein-protein interactions

Assessment of expression levels of *TaBES1s*

In Fig. 6, *TaBES1s* expression levels could be detected during wheat grain development. The expression levels of *TaBES1-6* and *TaBES1-7* showed no significant variation across the four varieties. The expression level of *TaBES1-1* gradually decreased during wheat grain development among the four varieties. The expression

trend of *TaBES1-5* was comparable among the four varieties, initially decreasing before subsequently increasing. For *TaBES1-2*, the expression trends in Yaomai36 and Yaomai30 were alike, which decreased first and then increased, while in Pinyu8155, the expression level initially rose before declining, and in Yaomai30, it exhibited a steady decrease. The expression levels of *TaBES1-3* in Yaomai36 and Yaomai30 showed no significant change. In Pinyu8175, the expression level was increased gradually, while in Pinyu8155, there was a gradual decrease. The expression levels of *TaBES1-4* in Yaomai36 and Yaomai30 were both gradually decreasing. In Pinyu8175, the expression level first declined and then rose, while in Pinyu8155, there were no significant changes. This indicates that *TaBES1-1/3/4* was specifically expressed in four varieties.

Transcriptional autoactivation and protein-protein interaction analysis of *TaBES1s*

All *TaBES1s* were evaluated for autoactivating activity in yeast. None of the *TaBES1s* possessed self-activating activity (Fig. 7A). As shown in Fig. 7B and C, and Table S9, no interaction partners were predicted for *TaBES1-3*. In contrast, three proteins (A0A3B6AUH8, A0A3B6C3H6, and A0A3B6DDH5) associated with *TaBES1-1* were predicted to share five common interaction partners (TraesCS1B03G0985600, TraesCS3A03G0312800, TraesCS3D03G0288900, TraesCS4A03G090110, and TraesCS5A03G0644800). Among these, three were classified as Uncharacterized proteins, while two were identified as members of the protein kinase superfamily. Two proteins (A0A3B6IKB1 and A0A3B6JE43) associated with *TaBES1-4* were predicted to share five common interaction partners (TraesCS2D03G0349000, TraesCS2A03G0256000, TraesCS2A03G0327100,

TraesCS2B03G0447100, and TraesCS2B03G0351500). Among these, three are 4- α -glucan transferases, and the other two are 4- α -glucan transferase DPE2. The predicted interactions were inferred from gene neighborhood and text mining analyses, with confidence scores of 0.945 (Table S9).

Discussion

BES1 is a plant-specific transcription factor essential for regulating plant growth and conferring stress resistance [28]. We identified 19 *TaBES1s* in wheat based on the genome-wide search method, which exhibit the typical domain (PF05687) (Fig. 1). These *TaBES1s* were further partitioned into four groups (Fig. 2A), consistent with the classification results obtained from studies on Arabidopsis [8] and rice [9]. Members of the same subfamily share similar exon–intron architectures and conserved motifs, implying functional coherence (Fig. 2A). There are differences in motif composition and gene structure between different subfamilies, which may indicate functional differences among them. For instance, subfamily I members are predominantly expressed in anthers and are almost not expressed during endosperm development or under abiotic stress (Fig. 5). In contrast, subfamily IV members exhibit expression in stamens, endosperm, and under abiotic stresses such as salt stress and heat stress (Fig. 5). Compared with the collinearity detected between Arabidopsis and wheat, the wheat–rice comparison revealed a substantially larger number of collinear gene pairs (Fig. 2C). This likely reflects the closer phylogenetic relationship between wheat and rice, both of which are members of the Poaceae monocots. Tandem replication and fragment replication are one of the primary mechanisms for the generation and expansion of plant gene families [29]. The collinear gene pair of *TaBES1s* was major derived from fragment replication, demonstrating that this process constitutes the predominant mechanism underlying *TaBES1* family expansion (Figure S1A). In contrast, our study identified 19 *TaBES1* genes based on the latest Chinese Spring genome information (IWGSC v2.1). To elucidate the source of these discrepancies, we compared our identified genes with those reported in previous studies. This comparative analysis revealed that four genes reported in previous studies were not identified in our analysis. This discrepancy likely stems from differences in the genome database versions, gene prediction methods, and software tools used. Furthermore, the study by DEZHOU W et al. primarily concentrated on the role of *TaBES1s* in stress resistance and anther development in wheat [2].

Predicting the functional roles of *TaBES1s* during grain development in wheat

Previous studies have demonstrated that *BES1s* transcription factors modulate rice grain development. Over-expression of *OsBES1-4* produces larger rice grains through increased cell elongation and proliferation [9]. Likewise, ectopic expression of *BES1/BZR1-5* markedly increases seed size and weight in both Arabidopsis and rice [30]. In this study, *OsBES1-4* (*LOC Os0lg10610.1*) clustered within the same subfamily as *TaBES1-3 A/3B/3D/6A/6B/6D/7A/7B/7D* (Fig. 2A). Additionally, *BES1/BZR1-5* (AT1G19350.1) and *TaBES1-1 A/1B/1D* were assigned to the same subfamily. Members of this subfamily are therefore expected to exhibit conserved biological functions. Collectively, *TaBES1-1 A/1B/1D/3A/3B/3D/6A/6B/6D/7A/7B/7D* are likely to play pivotal roles in wheat grain development. However, it should be noted that gene function is not entirely determined by its evolutionary relationship. Genes may undergo functional differentiation during evolution. Therefore, it is necessary to further verify the real function of *TaBES1s* in wheat, which can be achieved by gene editing and other methods. Related work is currently underway to elucidate the exact role of *TaBES1s* in wheat grain development.

Promoters also play a crucial role in gene expression patterns [31]. Although only a single *TaBES1* promoter displayed endosperm-specific activity (Fig. 4), this observation was corroborated in subsequent analyses (Fig. 5). Specifically, *TaBES1-2 A/2B/2D/4B/4D/5A/5B* were robustly expressed throughout the wheat grain development process (Figs. 5 and 6). However, integrative RNA-seq (Table 4) and qRT-PCR (Fig. 6) analyses widespread expression of *TaBES1s* throughout grain development, indicating a discrepancy between promoter element prediction and actual gene expression. This disparity likely arises from divergent genetic backgrounds: promoter annotation used the Chinese Spring, whereas RNA-seq data were generated from winter wheat accessions. The inherent genetic differences among these varieties may lead to differences in gene expression patterns. Moreover, the promoter may contain as-yet-unidentified cis-regulatory elements that govern grain development.

TaBES1s exhibited no intrinsic transcriptional self-activation, indicating that their regulatory function is contingent upon specific interactions with partner transcription factors or other regulatory proteins rather than autonomous function. Subsequently, the protein interaction of *TaBES1* was predicted. The results showed that the interacting proteins of *TaBES1-1* and *TaBES1-4* were not only involved in starch metabolism, but also may play a role in cell signal transduction. In addition, the interaction of *TaBES1-1* with some uncharacterized proteins suggests that they may have broader biological functions (Fig. 7B and C, Table S9). These findings lay the foundation for

further revealing the multiple biological functions of *TaBES1* in wheat. In the future, it is still necessary to verify these predicted interactions through experiments and explore their specific functions in wheat growth and development and environmental adaptation.

RNA-seq-based analysis of *TaBES1s* in various plant varieties

Pinyu8155 is categorized as dry-land wheat, whereas Yaomai36, Yaomai30, and Pinyu8175 are classified as irrigated wheat. Pinyu8155 exhibited significantly reduced grain diameter and thousand-grain weight compared with Yaomai30 (Figure S2). The grain length of Pinyu8155 was significantly lower than that of Pinyu8175 and Yaomai30. In contrast to Yaomai36, Yaomai30, and Pinyu8175, *TaBES1-3 A* was uniquely detected in Pinyu8155 grains (Table 4). Thus, *TaBES1-3 A* potentially represents a key regulator of grain filling in dryland wheat.

Wheat quality, a complex trait, commonly shows a negative correlation with yield [32]. Yaomai36 is classified as strong-gluten wheat, whereas Yaomai30, Pinyu8175, and Pinyu8155 are categorized as medium-gluten wheat (Table S7). Yaomai36 exhibited significantly reduced grain diameter and thousand-grain weight compared with the other three varieties, and the grain length was significantly lower than that of Pinyu8175 and Yaomai30 (Figure S2). Previous studies indicated that thousand-grain weight, a complex trait regulated by multiple genes, uses genetic parameters such as grain length and width as significant indicators for evaluating wheat grain quality [33]. In contrast to Yaomai36, *TaBES1-1D/3B/5B* was only expressed in Yaomai30, Pinyu8175, and Pinyu8155. Therefore, *TaBES1-1D/3B/5B* may only be involved in the grain development of medium-gluten wheat at the filling stage.

Given because of the significant genetic diversity among different wheat varieties and the differences in the expression patterns of *TaBES1s* genes related to grain development and quality traits, the effect of varieties on gene expression must be carefully considered, and the regulatory mechanisms behind these differences in future research.

Conclusions

In summary, *TaBES1s* were essential for plant development and stress responses. Previous studies have elucidated the role of *TaBES1s* in abiotic stress conditions and their role in rice grain development. This study further confirmed the potential function of *BES1s* in wheat seed development. We identified 19 *TaBES1* genes and analyzed the characteristics of gene family members, including conserved domains, phylogenetic relationships, gene structure, conserved motifs, collinearity,

promoter *cis*-acting elements, haplotypes, and expression patterns. Additionally, a comprehensive bioinformatics analysis and RNA-seq data revealed that *TaBES1s* exhibited distinct expression patterns in wheat grain development. These findings that *BES1s* play a regulatory role in wheat grain development provide a foundation for further understanding the molecular mechanisms underlying crop grain development. However, it must be pointed out that this study has not yet verified the exact role of *TaBES1s* in grain development at the functional level by means of experimental systems such as transgenic or gene editing. The conclusions are still based on the speculation of association analysis. Subsequent to this study, we will focus on the functional characterization of *TaBES1s* through gene-editing and overexpression approaches, in order to clarify their specific functions in wheat. This study provides a foundational framework for the in-depth dissection of the molecular mechanisms underlying wheat grain development.

Methods

Plant materials

The cultivation environment and sampling standards of the four experimental varieties (Yaomai36, Pinyu8175, Pinyu8155, and Yaomai30) were consistent with the published papers [34].

Genome-wide identification of *TaBES1s*

The genomics information of Chinese Spring wheat (IWGSC v2.1) was procured from the Joint Genome Institute website (<https://phytozome-next.jgi.doe.gov/v/>). The characteristic domain (PF05687) of the *BES1* gene family was acquired via the Basic Local Alignment Search Tool on the National Center for Biotechnology Information platform (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) [35] and subsequently downloaded from the Protein Families Database (PFAM) (<https://pfam.xfam.org/>) [36] to construct a Hidden Markov Model (HMM). The HMM search functionality within the HMMER3.0 software [37] was employed to retrieve *BES1* genes, and the results were subjected to strict filtering (E-value < 0.01). The transcripts of *TaBES1s* were screened utilizing the Phytozome website (<https://phytozome.jgi.doe.gov/pz/portal.html>) [38]. Redundant transcripts of the same gene were meticulously removed, with only the original transcripts retained. The protein sequences of *BES1* genes were extracted from the Chinese Spring wheat protein database using TBtools 1.115 software [44]. The raw data were refined to obtain high-quality protein sequences of *TaBES1s*. The Jalview 2.22.3.2 software was used to visualize the conserved protein domains of *TaBES1s* [39]. The fundamental physicochemical properties of *TaBES1s*, including amino acid length, molecular weight, and isoelectric point, were analyzed via the ExPASy-Prot Param

website [40] (<https://web.expasy.org/protparam/>). The chromosomal localization of *TaBES1s* was determined using the Phytozome website. The subcellular localization of BES1 genes was predicted through the Plant-mPLOC server website (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/#>) [41].

Phylogenetic relationships, chromosome distribution, and Col-linearity analysis of *TaBES1s*

The protein sequences of BES1 in rice and Arabidopsis were obtained from the Plant Transcription Factor Database (<http://planttfdb.gao-lab.org/index.php>). Multiple sequence alignments of BES1 in Arabidopsis thaliana, rice, and wheat were performed using MEGA 11 software. A phylogenetic tree was constructed using the neighbor-joining (NJ) method with a bootstrap value of 1000 [42]. The iTOL website [43] (<https://itol.embl.de>) was used for visual analysis of the results and subfamily classification. The TBtools 1.115 software was used to visualize the chromosomal localization of *TaBES1s* [26].

Genomics col-linearity relationships were investigated using the MCScanX algorithm implemented in TBtools 1.115, with subsequent visualization of syntenic relationships achieved through circos plot construction via the integrated toolkit [44]. NCBI BLAST was utilized to compare the CDS sequences of *TaBES1s* and to analyze the homology of gene pairs. The Ka/Ks_Calculator plug-in in TBtools 1.115 facilitated the calculation of the nonsynonymous substitution rate (Ka), synonymous substitution rate (Ks), and Ka/Ks ratio [45]. The col-linearity between Arabidopsis, rice, and wheat was visualized using TBtools 1.115.

Gene structure and conserved motif analysis of *TaBES1s*

The website of MEME analysis (<https://meme-suite.org/meme/tools/meme>) was employed to submit the BES1s protein sequences, resulting in the retrieval of a maximum of 10 motifs in the MEME file [46]. Visualization was achieved through the Gene Structure View plug-in in TBtools 1.115.

Cis-acting elements analysis of *TaBES1s*

The upstream 2000 bp CDS sequence of *TaBES1s* was obtained through TBtools. This sequence was submitted to the Plant-CARE website [47] (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for the prediction of promoter *cis*-elements.

Gene ontology enrichment and haplotype analysis of *TaBES1s*

A SNP analysis of *TaBES1s* was conducted using the agriGO v2.0 website (<http://systemsbiology.cau.edu.cn/agriGOv2/>) [48]. The haplotypes of *TaBES1s* were extracted based on the chromosomal positions of the

genes, utilizing whole genome re-sequencing data of wheat from Lufei (http://wheat.cau.edu.cn/WheatUnion/b_4/) [49]. The quantity of *TaBES1* haplotypes carrying genetic variation data was enumerated.

Gene expression analysis and qRT-PCR analysis of *TaBES1s*

Download FPKM values from various wheat organizations from the database website (<http://ipf.sustech.edu.cn/pub/wheatrna/>). Different samples of the filling stage in Yaomai36, Pinyu8175, Pinyu8155, and Yaomai30 were sent to Beijing Qingke Biotechnology Co., Ltd. (Beijing, China) for RNA-Seq. qRT-PCR analysis of *TaBES1s* was completed following the published papers [34].

Transcriptional self-activation analysis and interaction protein prediction of *TaBES1s*

The yeast two-hybrid system was used to confirm the transcriptional self-activation of *TaBES1s* [34]. The primer sequences used in the experiment are shown in Table S10. pGBKT7 blank vector and pGBKT7-PtrWOX13A as positive control and negative control, respectively [50]. The interaction proteins of *TaBES1s* were predicted using the UniProt Protein Database (<https://www.uniprot.org/>) and STRING (<https://cn.string-db.org/>).

Statistical analysis

The statistical significance of the data was assessed using Dunnett's test (SPSS 20.0). In this study, the differences between the two groups of data were evaluated for significance at the level of $p < 0.05$.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-11999-8>.

Supplementary Material 1.

Authors' contributions

Y.Z. and Y.X. performed the formal data collection and analyses, as well as preparing the original draft of the manuscript. Y.M., M.W., and X.L. participated in the RNA-seq analysis and qRT-PCR analysis. L.J., J.H., and D.Z. conceived and designed the study, X.M. and H.J. obtained funds and critically reviewed the final draft of the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding

The experimental design and implementation, RNA extraction, data preprocessing, and analysis, as well as interpretation, were supported by the Fundamental Research Program of Shanxi Province (202303021222075) and China Agriculture Research System (Wheat, CARS-03-54). The graduate students, Y.X. and Y.M. were supported by the Project of Science and Technology Innovation Fund of Shanxi Agricultural University (2023BQ40) and supported by the earmarked fund for Modern Agro-industry Technology Research System (2024CYJSTX02-22).

Data availability

The genomics information of Chinese Spring wheat (IWGSC v2.1) was procured from the Joint Genome Institute website (<https://phytozome-nex>

t.jgi.doe.gov/). The protein sequences of BES1 in rice and Arabidopsis were obtained from the Plant Transcription Factor Database (<http://planttfdb.gao-lab.org/index.php>). The haplotypes of TaBES1s were extracted based on the chromosomal positions of the genes, utilizing whole genome re-sequencing data of wheat from Lufei (http://wheat.cau.edu.cn/WheatUnion/b_4/). Download FPKM values from various wheat organizations from the database website (<http://ipf.sustech.edu.cn/pub/wheatrna/>). Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 25 March 2025 / Accepted: 11 August 2025

Published online: 26 September 2025

References

- Xuejing C, Weifeng M, Fanwei Z, Yongjuan C, Zonghuan M, Juan M, Baihong C. Grape BES1 transcription factor gene VvBES1-3 confers salt tolerance in Transgenic Arabidopsis. *Gene*. 2023;854:147059–147059.
- Nemanja MNT, Derui V, Eugenia L, Yanhai R. Brassinosteroids: multidimensional regulators of plant growth, development, and stress responses. *Plant Cell*. 2020;32(2):295–318.
- Alfredo K, Yanhai Y. Updates on BES1/BZR1 regulatory networks coordinating plant growth and stress responses. *Front Plant Sci*. 2020;11:617162–617162.
- Bulgakov VP, Avramenko TV. Linking brassinosteroid and ABA signaling in the context of stress acclimation. *Int J Mol Sci*. 2020;21(14):5108.
- Yin Y, Wang Z-Y, Mora-Garcia S, Li J, Yoshida S, Asami T, Chory J. BES1 accumulates in the nucleus in response to brassinosteroids to regulate gene expression and promote stem elongation. *Cell*. 2002;109(2):181–91.
- Yanhai Y, Dionne V, Yi T, Shigeo Y, Tadao A, Joanne C. A new class of transcription factors mediates brassinosteroid-regulated gene expression in Arabidopsis. *Cell*. 2005;120(2):249–59.
- Teni B. Putative nuclear localization signals (NLS) in protein transcription factors. *J Cell Biochem*. 1994;55(1):32–58.
- Xiaofei Y, Lei L, Jaroslav Z, Maneesha A, Huaxun Y, Andrew F, Hongqing G, Sarah A, Srinivas A, Peng L, et al. A brassinosteroid transcriptional network revealed by genome-wide identification of BES1 target genes in Arabidopsis Thaliana. *Plant Journal: Cell Mol Biology*. 2011;65(4):634–46.
- Mingxing C, Huanran Y, Ruihua W, Wei W, Licheng Z, Fengfeng F, Shaoqing L. Identification and characterization of BES1 genes involved in grain size development of *Oryza sativa* L. *Int J Biol Macromol*. 2023;253(P6):127327–127327.
- Dezhou W, Jinghong Z, Shan L, Weiwei W, Qing L, Xiaocong H, Zhaofeng F, Ting L, Yue S, Chunman G, et al. BRI1 EMS SUPPRESSOR1 genes regulate abiotic stress and anther development in wheat (*Triticum aestivum* L.). *Front Plant Sci*. 2023;14:1219856–1219856.
- Yu H, Feng W, Sun F, Zhang Y, Qu J, Liu B, Lu F, Yang L, Fu F, Li W. Cloning and characterization of BES1/BZR1 transcription factor genes in maize. *Plant Growth Regul*. 2018;86(2):235–49.
- Si M, Tingting J, Meiting L, Shihui L, Yongqiang T, Lihong G. Genome-Wide identification, structural, and gene expression analysis of BRI1-EMS-Suppressor 1 transcription factor family in *cucumis sativus*. *Front Genet*. 2020;11:583996–583996.
- Li H, Zou T, Chen S, Zhong M. Genome-wide identification, characterization and expression analysis of the DUF668 gene family in tomato. *PeerJ*. 2024;12:e17537–17537.
- Zhang T, Xu P, Wang W, Wang S, Caruana JC, Yang HQ, Lian H. Arabidopsis G-Protein β subunit AGB1 interacts with BES1 to regulate brassinosteroid signaling and cell elongation. *Front Plant Sci*. 2018;8:2225.
- Yuanquan Z, Beibei L, Yunyan X, Heng L, Shanshan L, Dajian Z, Zhiwei M, Siyi G, Chunhong Y, Yuxiang W, et al. The Cyclophilin CYP20-2 modulates the conformation of BRASSINAZOLE-RESISTANT1, which binds the promoter of FLOWERING LOCUS D to regulate flowering in Arabidopsis. *Plant Cell*. 2013;25(7):2504–21.
- Youn SS, Hayung C, Young KS, Hee NK. BRI1-EMS-suppressor 1 gain-of-function mutant shows higher susceptibility to necrotrophic fungal infection. *Biochem Biophys Res Commun*. 2016;470(4):864–9.
- Hojin R, Hyunwoo C, Wonsil B, Ildoo H. Control of early seedling development by BES1/TPL/HDA19-mediated epigenetic regulation of ABI3. *Nat Commun*. 2014;5(1):4138.
- Zhao X, Dou L, Gong Z, Wang X, Mao T. BES1 hinders ABCISIC ACID INSENSITIVE5 and promotes seed germination in Arabidopsis. *New Phytol*. 2019;221(2):908–18.
- Wenqi F, Hongwanjun Z, Yang C, Yuan L, Yiran Z, Fuai S, Qingqing Y, Xuecai Z, Yuanyuan Z, Yingge W. Maize ZmBES1/BZR1-1 transcription factor negatively regulates drought tolerance. *Plant Physiol Biochemistry: PPB*. 2023;205:108188–108188.
- Yan Q, Wang L, Li X. GmBEHL1, a BES1/BZR1 family protein, negatively regulates soybean nodulation. *Sci Rep*. 2018;8(1):1–12.
- Xiaofang L, Caide Z, Yuqi G, Yao X, Shujin W, Chaoshuo L, Yinpeng X, Pengxiang C, Peizhi Y, Li Y, et al. A multifaceted module of BRI1 ETHYLMETHANE SULFONATE SUPPRESSOR1 (BES1)-MYB88 in growth and stress tolerance of Apple. *Plant Physiol*. 2021;185(4):1903–23.
- Rudi SGB, Anna-Maria A, Robin B-O, Boulos BCLBJ, Forrest C, Jan C, Masaru D, Beat I. A workshop report on wheat genome sequencing: international genome research on wheat consortium. *Genetics*. 2004;168(2):1087–96.
- Li S, Zhang C, Li J, Yan L, Wang N, Xia L. Present and future prospects for wheat improvement through genome editing and advanced technologies. *Plant Commun*. 2021;2(4):100211.
- Wang Z, Yin Y, He M, Zhang Y, Lu S, Li Q, Shi S. Allocation of photosynthates and grain growth of two wheat cultivars with different potential grain growth in response to Pre- and Post-anthesis shading. *J Agron Crop Sci*. 2003;189(5):280–5.
- Juan G, Liquan L, Qian L, Yi Z, Yan L, Li Z, Xuejun L. TaGW2-6A allelic variation contributes to grain size possibly by regulating the expression of cytokinins and starch-related genes in wheat. *Planta*. 2017;246(6):1153–63.
- Deding S, Wei X, Ling W, Wang L, Yuan S, Yudong L, Zhengguo L. Genome-wide identification, characterization and expression analysis of BES1 gene family in tomato. *BMC Plant Biol*. 2021;21(1):161–161.
- Li Q, Guo L, Wang H, Zhang Y, Fan C, Shen Y. In Silico genome-wide identification and comprehensive characterization of the BES1 gene family in soybean. *Heliyon*. 2019;5(6):e01868.
- Zhenting S, Xingzhou L, Weidong Z, Huan L, Xiugui C, Yan L, Wuwei Y, Zujun Y. Molecular traits and functional exploration of BES1 gene family in plants. *Int J Mol Sci*. 2022;23(8):4242–4242.
- Cannon SB, Mitra A, Baumgarten A, Young ND, May G. The roles of segmental and tandem gene duplication in the evolution of large gene families in Arabidopsis Thaliana. *BMC Plant Biol*. 2004;4:1–21.
- Fuai S, Lei D, Wenqi F, Yang C, Fengzhong L, Qingqing Y, Wanchen L, Yanli L, Nitzan S, Fengling F, et al. Maize transcription factor ZmBES1/BZR1-5 positively regulates kernel size. *J Exp Bot*. 2020;72(5):1714–26.
- Yang J, Zhang B, Gu G, Yuan J, Shen S, Jin L, Lin Z, Lin J, Xie X. Genome-wide identification and expression analysis of the R2R3-MYB gene family in tobacco (*Nicotiana tabacum* L.). *BMC Genomics*. 2022;23(1):432.
- Shichen Z, K FA, Guorong Z, Byron E, Rebecca R, Jesse P. Accelerating wheat breeding for end-use quality through association mapping and multivariate genomic prediction. *Plant Genome*. 2021;14(3):e20164–20164.
- Campbell KG, Bergman CJ, Gualberto DG, Anderson JA, Giroux MJ, Hareland G, Fulcher RG, Sorrells ME, Finney PL. Quantitative trait loci associated with kernel traits in a soft wheat cross. *Crop Sci*. 1999;39(4):1184–95.
- Zhang Y, Xu Y, Mao Y, Tan X, Tian Y, Ma X, Ji H, Zhang D. Genome-Wide identification and expression analysis of NF-YA gene family in the filling stage of wheat (*Triticum aestivum* L.). *Int J Mol Sci*. 2025;26:133.
- McGinnis S, Madden TL. BLAST: at the core of a powerful and diverse set of sequence analysis tools. *Nucleic Acids Res*. 2004;32(suppl2):W20–5.
- Finn RD, Mistry J, Schuster-Böckler B, Griffiths-Jones S, Hollich V, Lassmann T, Moxon S, Marshall M, Khanna A, Durbin R. Pfam: clans, web tools and services. *Nucleic Acids Res*. 2006;34(suppl1):D247–51.
- Finn RD, Clements J, Eddy SR. HMMER web server: interactive sequence similarity searching. *Nucleic Acids Res*. 2011;39(suppl):W29–37.
- Goodstein DM, Shu S, Howson R, Neupane R, Hayes RD, Fazo J, Mitros T, Dirks W, Hellsten U, Putnam N. Phytozome: a comparative platform for green plant genomics. *Nucleic Acids Res*. 2012;40(D1):D1178–86.

39. Waterhouse AM, Procter JB, Martin DM, Clamp M, Barton GJ. Jalview version 2—a multiple sequence alignment editor and analysis workbench. *Bioinformatics*. 2009;25(9):1189–91.
40. Gasteiger E, Gattiker A, Hoogland C, Ivanyi I, Appel RD, Bairoch A. ExPASy: the proteomics server for in-depth protein knowledge and analysis. *Nucleic Acids Res*. 2003;31(13):3784–8.
41. Chou K-C, Shen H-B. Plant-mPLOC: a top-down strategy to augment the power for predicting plant protein subcellular localization. *PLoS ONE*. 2010;5(6):e11335.
42. Hall BG. Building phylogenetic trees from molecular data with MEGA. *Mol Biol Evol*. 2013;30(5):1229–35.
43. Letunic I, Bork P. Interactive tree of life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Res*. 2021;49(W1):W293–6.
44. Wang Y, Tang H, DeBarry JD, Tan X, Li J, Wang X, Lee T-h, Jin H, Marler B, Guo H. MCSscanX: a toolkit for detection and evolutionary analysis of gene synteny and collinearity. *Nucleic Acids Res*. 2012;40(7):e49–49.
45. Zhang Z, Li J, Zhao X-Q, Wang J, Wong GK-S, Yu J. KaKs_Calculator: calculating Ka and Ks through model selection and model averaging. *Genomics Proteom Bioinf*. 2006;4(4):259–63.
46. Bailey TL, Johnson J, Grant CE, Noble WS. The MEME suite. *Nucleic Acids Res*. 2015;43(W1):W39–49.
47. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rambauts S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in Silico analysis of promoter sequences. *Nucleic Acids Res*. 2002;30(1):325–7.
48. Tian T, Liu Y, Yan H, You Q, Yi X, Du Z, Xu W, Su Z. AgriGO v2.0: a GO analysis toolkit for the agricultural community, 2017 update. *Nucleic Acids Res*. 2017;45(W1):W122–9.
49. Zhou Y, Zhao X, Li Y, Xu J, Bi A, Kang L, Xu D, Chen H, Wang Y, Wang Y. -g: triticum population sequencing provides insights into wheat adaptation. *Nat Genet*. 2020;52(12):1412–22.
50. Yang Z, Yingying L, Xueying W, Ruiqi W, Xuebing C, Shuang W, Hairong W, Zhigang W. PtrWOX13A promotes wood formation and bioactive gibberellins biosynthesis in *Populus trichocarpa*. *Front Plant Sci*. 2022;13:835035.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.